University of Athens
Department of Informatics and Telecommunications

Operating Systems (K22) / Period 2020-2021
2nd Assignment

In this assignment we will examine the behavior of applications with respect to the replacement algorithms. The application will simulate the behavior of a memory management system based on a real reference trace. You are given two reference files [1], [2]. Each file contains a number of lines in which hexadecimal memory references are presented followed by the indicators R or W that specify whether it is a write or a read. The simulator will read the file and will execute the virtual memory mechanism based on a hashed page table. The simulator must know which pages have been loaded into main memory. When a reference arises from the reference trace the simulator must check whether the specific page is in main memory. If this does not hold and the memory is already full it must decide which page to remove (page replacement). If the page that is to be removed has been modified it must be "saved" to the disk. When a new page is loaded the page table must also be updated accordingly. All pages and frames have a size of 4096 bytes. The LRU and Second Chance algorithms must be implemented. When the need arises for writing to the disk (or reading) the event must be recorded in appropriate counters of the simulation program. The program will alternately read sets of references (of size q) from each trace file so that main memory contains pages from both of the 2 processes.

The program will accept as arguments the replacement algorithm, the number of frames of main memory and the number of references q. Also, the maximum number of references (max) that will be examined from the trace files may be specified. Upon completion of the simulation statistics must be displayed such as number of writes/reads from the disk, page faults, entries examined from the reference trace file, number of memory frames.

Due Date: 14/01/2021
Submission method: submission on eclass, a tar file must be submitted containing all the relevant files: source and header files, makefile, etc.
Accompanying material: 2-page documentation that explains the program and gives brief technical details.
Implementation: the assignment is individual, it must be implemented in the C/C++ language.
The assignment will be examined on Linux systems of the Department according to a schedule that will be announced right after the due date.

[1] http://eclass.uoa.gr/modules/document/file.php/D244/CW_material/bzip.trace.gz
[2] http://eclass.uoa.gr/modules/document/file.php/D244/CW_material/gcc.trace.gz